AMERICAS TECHNOLOGY: SOFTWARE

Revisiting Moats Part I: Exploring AI Steel Men Arguments

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Relative to prior Software sector corrections, the 2026 YTD one is squarely anchored to terminal value debates: instead of analyzing a change in near-term demand trendline or the mathematical impact of higher interest rates, the market is questioning software moats and business models. While this debate is centered primarily on the application software companies; to a lesser extent, investors are also questioning whether the infrastructure/security stack could be impacted, and the ROI tied to hyperscaler capex. In this context, we offer thoughts on the seven primary bear cases we hear on Software, roughly order each bear case via a risk score of 1 (low) to 5 (high), and arrange on an axis of narrowly defined to app software vs. broadly defined (Exhibit 1). Our goal is to revisit this topic of moats over the next several months to better understand barrier to entry, product differentiation and pricing power. Our key observation over the past four weeks has been how quickly the agentic technology ecosystem is evolving, which makes it challenging to assess terminal values and thus put a floor to valuations — nonetheless, we offer signals to watch for stabilization in our 2/8 note here. (In short, we look for proof that domain experience drives higher quality agentic outcomes, and stable to improving fundamentals through earnings). In the meanwhile, we highlight business models with a clear architectural moat that extends beyond the application layer, including the following Buy rated names: CRWD, GWRE, KVYO, MDB, NET, NOW, RBRK, SHOP, SNOW.

Additional reading:

- Generative AI Part XI: Agentic AI expands the App Software TAM (June 2025)
- Assuming Software Sector Coverage into the Decade of Agentic Workflow (Jan 2026)
- Vertical Software Well-Positioned to Outperform as Generative AI Evolves (Jan 2026)
- Signals that could lead to stabilization & feedback from investor conversations (Feb 2026)
- Addressing AI Disruption Debate Across SMID Coverage (Feb 2026)
- Revisiting Key Debates: Vertical Software a Bright Spot in 4Q Industry Data (Feb 2026)

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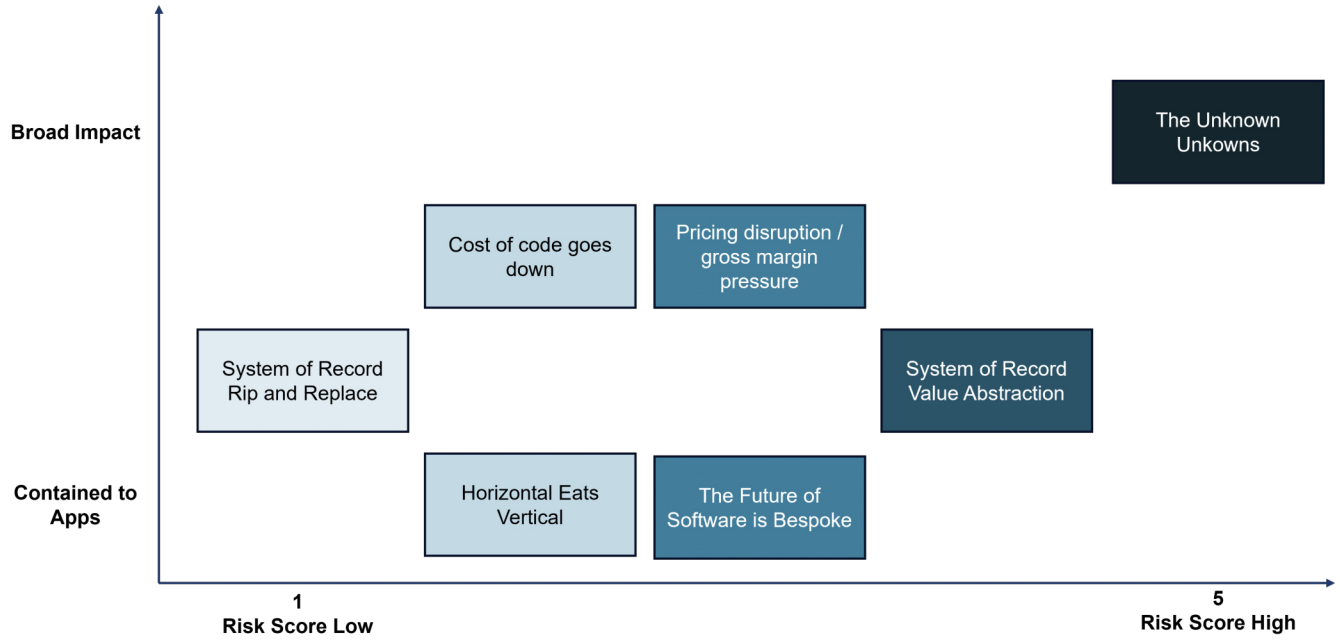
Exhibit 1: We order seven bear case debates on a scale of "Straw Men" to "Steel Men", and Narrow vs. Broad impact



Source: Goldman Sachs Global Investment Research

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A. System of Record "rip and replace" — risk score 1

The bear case: New competition comes up with new ways to solve the system of record problem for storing foundational business data, catalyzing a "rip and replace" movement making today's SoR-centric software companies obsolete.

GS View: *We see relatively low risk of this happening because GenAI is designed to be an analytical and generative engine, not a transactional engine, and assign a risk score of "1". Being able to underwrite core systems of record staying intact (even in a world where they enable a new set of competition or where parts of their stack like the user interface are disintermediated) allows us to underwrite a terminal value that is not zero, but rather closer to where most SoR companies are currently trading.*

Key Question:

- **What is the value of the System of Record within the stack?** If Exhibit 2 holds true in a reasonable bear case scenario, the value of app software within the technology stack will effectively compress to the value of the System of Record, while the profit pools beyond the SoR could largely be captured by new competition. Existing software incumbents today will become intelligent data storage layers - but even then, these layers do not have a terminal value of zero and thus can help underwrite valuation support.

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A meaningful part of the Software TAM has historically been captured by Systems of Records (SoRs). These systems such as ERP (enterprise resource planning), CRM (customer relationship management), and HR (human resources) are the digital backbone of the business, storing the foundational data upon which the business operates.

Why AI currently relies on existing systems of record: AI technologies function primarily as a powerful intelligent layer rather than a foundational replacement: they are reliant on systems of record to maximize their own value. AI models, particularly in an Enterprise context, require vast amounts of high-quality, structured, and historically accurate data for training and operation. The existing systems of record are the reservoirs of this data. Existing SoR companies including SAP, Salesforce, Oracle, and Workday have spent decades building robust processes for data validation, governance, and compliance. An AI application for financial planning & analysis, for example, must pull accurate historical figures from a trusted ledger that has been meticulously maintained and audited. In another example, AI analyzing a sales call transcript can provide meaningful insight if it can connect that conversation with the customer's data currently stored in CRM or ERP systems. There may be commoditization risk for the Agent itself, but the data and historical context sitting in the ledger cannot be extracted holistically and in a way that captures ongoing data updates. This is a key strength of existing systems of record, which are engineering for high-volume, mission-critical data processing to ensure every transaction is recorded accurately. As data accumulates in the existing Systems of Record, it attracts applications and services to it that may further necessitate buy-in for that system's functionality: an example of the Data Gravity phenomenon that long precedes the AI era. AI technologies are currently designed to be an analytical and generative engine, not a transactional engine. The value of the

intelligence layer is then increasingly determined by the gravity of the underlying record layer.

The risk of replacement...: While today's AI still relies on existing Systems of Record, the risk would be if new companies also reinvent the SoR layer—driven both by financial benefits (lower total cost of ownership and stronger ROI) and by technical advantages (a more modern, scalable architecture). From a financial standpoint, legacy ERP programs are widely associated with high cost and long, complex delivery cycles. For example, SAP is currently moving customers through its S/4HANA cloud upgrade cycle, and migrations for large enterprises are often described as multi-phase efforts that can span 18–36 months.

...and defending against it: The best way in our view for incumbents to mitigate displacement risk is to evolve from being largely passive ledgers focused on storing data to becoming active platforms that support automation and more intelligent, real-time operations:

- **Shift from passive record keeping to active reasoning:** Traditional Systems of Record have largely functioned as passive ledgers that store a company's most important data. As agentic technologies mature, these platforms are in various stages of evolving from data repositories to systems of reason. In practice, the System of Record should act as an intelligent foundation that actively enables automation and decision execution, rather than simply holding information for downstream tools to use.
- **AI powered versus AI native:** Most enterprises today implement AI powered solutions, where AI is added onto existing applications while the underlying architecture stays largely the same. This approach can constrain agentic automation because it often depends on static data pipelines and rigid structures. As new AI capabilities are introduced, organizations typically need additional integrations, which can increase complexity and cost while still limiting end to end functionality. By contrast, AI native architectures are designed around intelligence and automation from the start. Data access and decision-making are embedded into the core platform rather than layered on top of the existing architecture, enabling continuous feedback loops that help models and workflows improve over time. These systems commonly rely on agents, orchestration layers, and near real time data flows to support adaptive, automated operations. In our view, the question then becomes whether existing SoRs will be able to leverage existing architectural moats and innovate at a pace that insulates profit pools from AI-native new entrants. We are already seeing signs of SoRs meaningfully reinventing themselves, such as Salesforce's 2024 replatforming and Workday announcing a 180-degree transformation from closed to open.

We believe growth-equity firm Avenir details these points well: 1) there are often cheaper software options available, yet more expensive players continue to win and still only represent a small portion of total employee expense; 2) larger SaaS providers perform better on feature completeness, ecosystem partnerships, employee familiarity, security and scalability; 3) SaaS providers have the same access to foundation models as other companies (eg. OpenAI & Anthropic are both seeing strong growth from their API businesses, partially driven by incumbent SaaS providers); and 4) early enterprise AI

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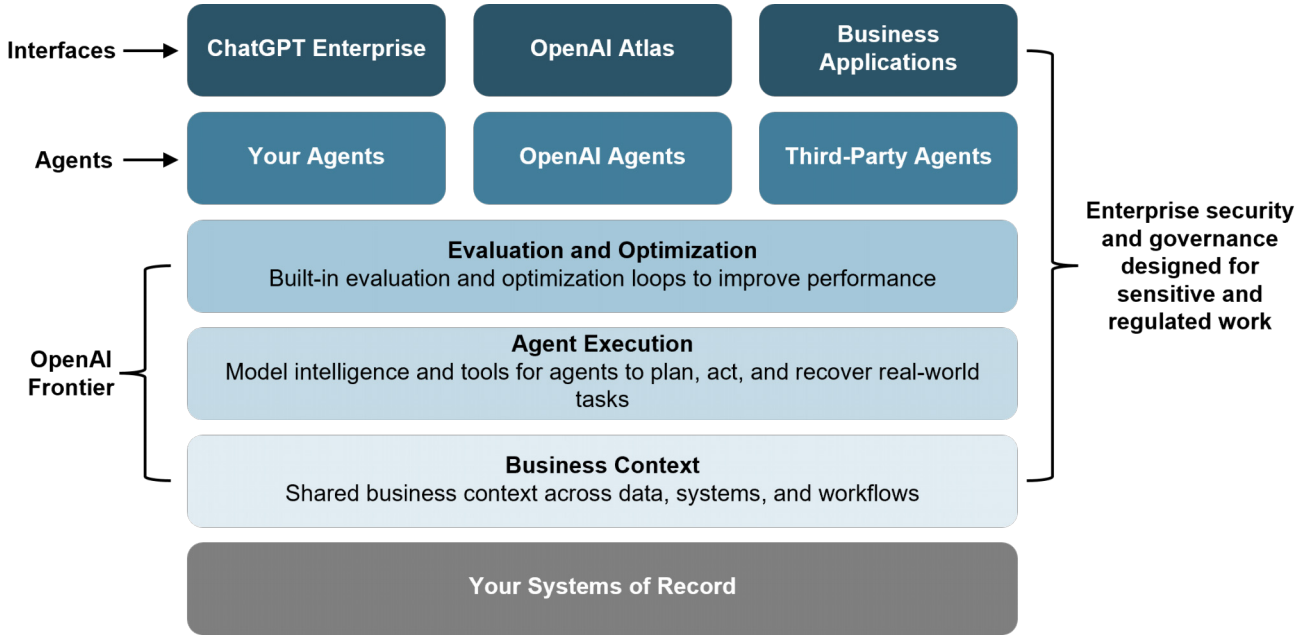
use-cases in customer support or coding have required less context and agency vs. future use cases which will result in AI depending more on Systems of Record to both provide AI the context it needs and allow AI to work alongside humans. This also ties into our sections on Vibe Coding (D) and Bespoke Software (E).

A critical determinant of the future technology stack is data accessibility, as the prevailing corporate stance on data sharing will dictate the end-state role of the System of Record. If enterprises continue to 'cordon-off' data advantages within existing applications (a trend exemplified by Salesforce's May 2025 change to its Slack API terms, prohibiting LLM training and bulk export of Slack messages and file data), then the OpenAI Frontier model (Exhibit 2), which positions SoRs as the foundational layer beneath intelligence, becomes the most probable framework. This data siloing has a direct consequence: while top AI models will still permeate bespoke use cases, their broader application will be narrowed in scope, as they will be cut off from the rich, contextualized data held within the SoR. The balancing factor is customer tolerance: we think the steady state is not full lock down, but controlled interoperability (real-time retrieval, scoped permissions) that protects proprietary data while enabling enterprise-grade AI integrations. That brings the debate back to innovation: whether SoRs can deliver AI-native workflows natively in product while still supporting integration patterns enterprises require; and we dig into that dynamic in the next section.

Exhibit 2: OpenAI's "Frontier" exhibit suggests existing SORs will sit below multiple layers of intelligence but even in this vision the SoR still exists; the risk is that the value gets abstracted away from the SOR, not that the SOR gets replaced
OpenAI Frontier architecture



Source: OpenAI

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B. Value accrues away from Systems of Record and towards the Agentic Operating System — risk score 4

The bear case: New AI tools sit on top of the existing Software stack and capture incremental value. While SoRs remain necessary for data storage and compliance, they increasingly function as infrastructure rather than primary value creators. Today's application software leaders have limited exposure to growth and outsized exposure to seat count reductions or commoditization.

GS View: The risk from AI in our view is less on rip and replace, but more on value abstraction. The biggest advantage that incumbents have is their domain experience and context. We expect select incumbents will be able to prove to customers that this domain experience drives better AI outcomes, provided they have been thoughtful about cleaning up tech debt/keeping their stack dynamic and have innovated as fast followers. The incumbents are not static: many have world class engineering teams that benefit from the same AI productivity gains as start ups, founder involvement, and strong cash balances to respond to new competitive threats. However, not all companies are well positioned to execute in a period of intensifying competition, and we assign this bear case a risk score of "4".

As agentic systems mature, intelligence moves outside the SoR and into an orchestration layer that can reason across multiple systems, invoke APIs, and execute workflows autonomously. Agents combine models, workflows, and APIs to perform tasks directly, with value increasingly concentrated in the platform layer that processes data and coordinates actions. Agents can query, write to, and reconcile data across multiple SoRs without users interacting with those systems directly. In this architecture, the SoR's role is to reliably store and expose data, while decision making, prioritization, and workflow logic reside elsewhere. This weakens traditional SoR moats tied to UI, workflow ownership, and user habituation. The SoR becomes progressively decoupled from how work is done. New winners that capture new budget are companies that control agent platforms, orchestration layers, and domain-specific AI systems that sit above SoRs. Tools like Sierra would sit on top of Salesforce, while Anthropic Cowork would sit on top of Microsoft, with incremental value increasingly captured by Sierra and Anthropic. The SoR becomes a commoditized component of a broader AI stack and increasingly competes on reliability, compliance, and cost. We believe the competitive noise on this bear case is particularly loud because there were a set of application companies funded and/or scaled in the 2020/2021 low-rates investment cycle that have relatively weaker moats and are more easily disintermediated, fueling a narrative of broader disintermediation.

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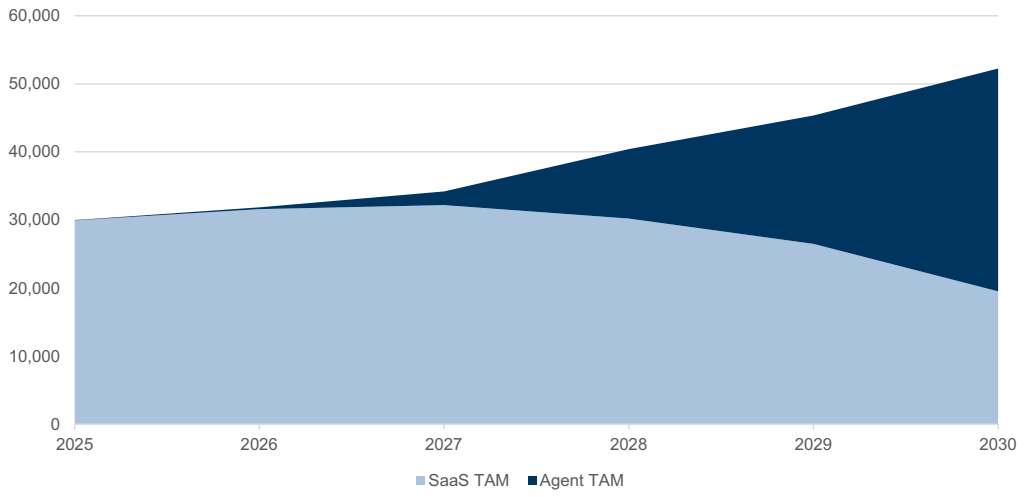
Exhibit 3: Our TAM math on the Customer Service subsector shows that the TAM is ultimately bigger with a shift from SaaS to Agents because of realized productivity gains; today's Software leaders need to capture part of the upside to avoid being overly exposed to the downside

GS math on how the customer service software TAM evolves over time



Source: Gartner, Goldman Sachs Global Investment Research

In our view, the biggest advantage that incumbents have is their domain experience and context. We expect select incumbents will be able to prove to customers that their domain experience drives better AI outcomes, provided they have been thoughtful about cleaning up tech debt/keeping their stack dynamic and have innovated as fast followers.

- **Microsoft's perspective**: While customers own their own data, Microsoft highlights the differentiation customers achieve by staying in the Microsoft ecosystem as it reduces latency, ensures data is current, and provides LLMs more context. For example, an agent running in Excel outside of the Microsoft ecosystem will likely struggle to pull context from other sources (e.g. Outlook); however, Copilot can. This enables more seamless integration of AI into everyday workflows. Additionally, in periods of platform change, it continues to be expensive and risky to move data and customers consistently underestimate the friction, cost, and breakage involved in egressing data at scale. This has even resulted in Microsoft seeing customer repatriation from third-party data management solutions back into its native stack.
- **HubSpot's perspective:** With the launch of its Agentic Customer Platform in February, HubSpot argues that AI is underperforming in business use cases because it lacks context given that AI models are trained on the internet and therefore do not know an individual business. Context in this case is defined as customer history and intent; why past decisions were made; how teams prioritize, escalate, and collaborate; and what works for similar companies in the industry. This has resulted in frustration from knowledge workers, particularly as many see significant value from AI in their personal lives. Context is spread across fragmented CRM records, emails, calls, chats, internal tools, outside of systems, and lost when employees leave. AI tools require users to manually re-teach this context, creating inconsistent

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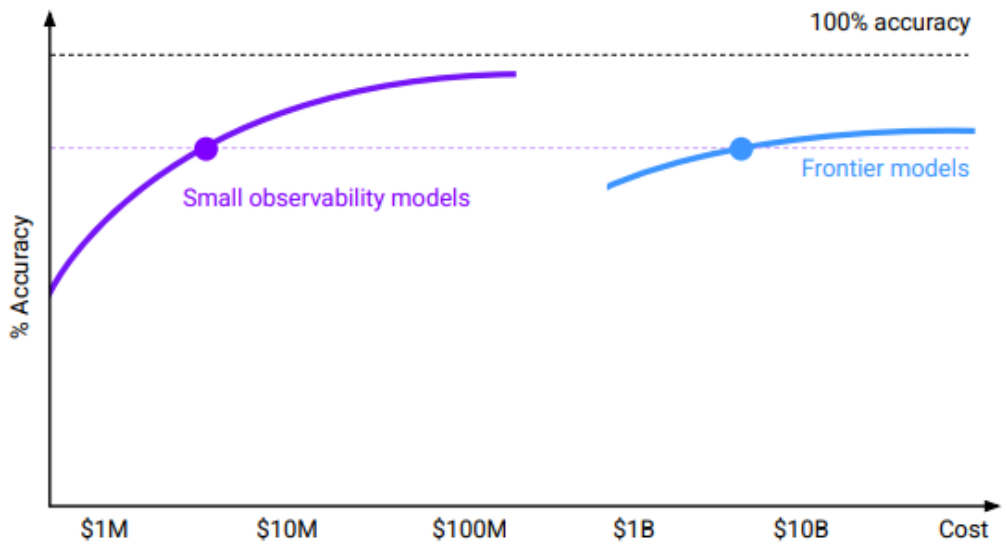
results. A system of record helps aggregate this and AI built into these applications will allow humans to work alongside agents — each with the same underlying context. HubSpot's tech stack is particularly well positioned for this transition given its focus on centralized R&D (via the primary colors) and its minimal tech debt, as the vast majority of its innovation has been organic.

- **Datadog's perspective:** Datadog is an observability infrastructure provider, and in some ways can be viewed as providing an observability application that includes a best in class user interface to engineering teams. At Datadog's 2/12 analyst day, Datadog's cofounders elegantly showed the importance of context by illustrating that its SLMs trained on its internal data are able to deliver better accuracy at lower cost than frontier models because of its domain experience. We think similar illustrations from app software companies would also prove this point of context being valuable.

Exhibit 4: Datadog's cofounders elegantly showed the importance of context: SLMs trained on its internal data deliver better accuracy at lowre cost than frontier models because of its domain experience
An illustrative example from Datadog's 2026 investor day



Source: Datadog

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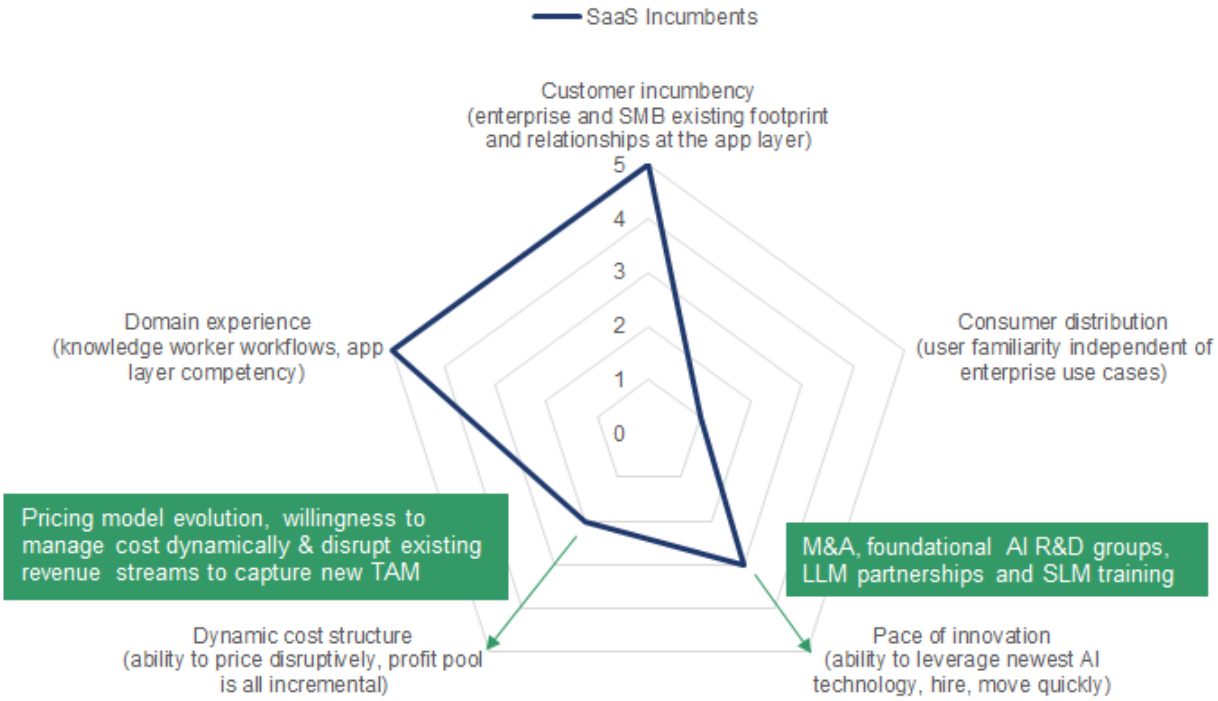
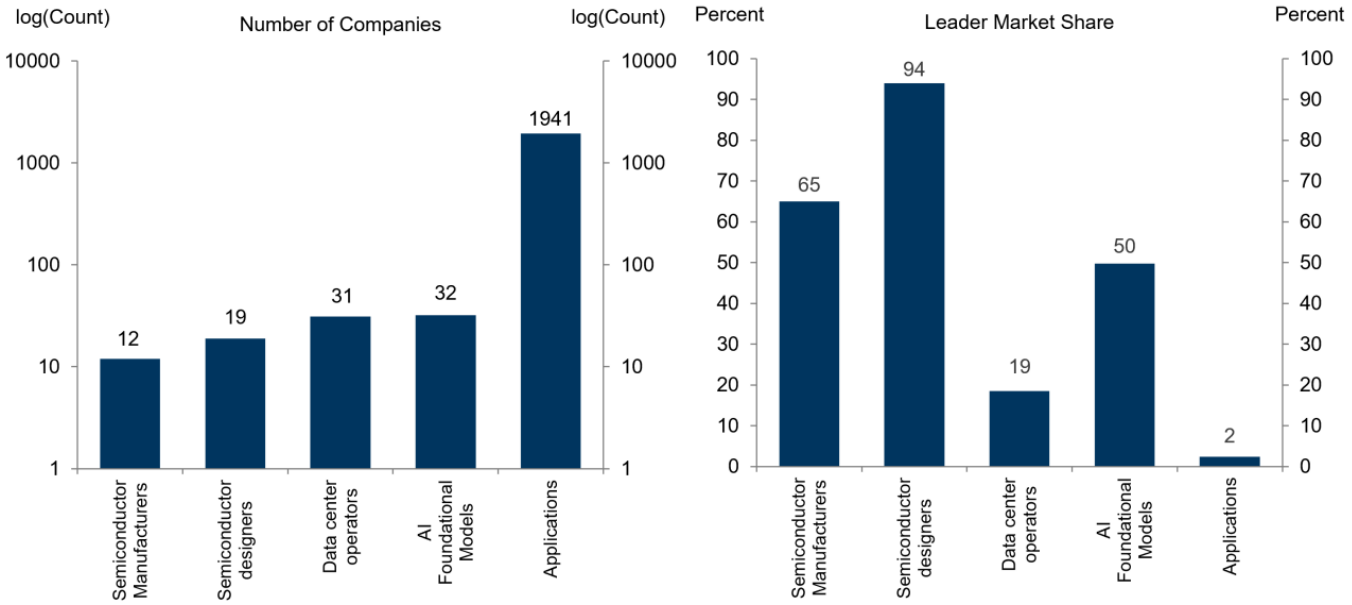
Exhibit 5: App companies can improve their positioning by innovating organically and inorganically



Source: Goldman Sachs Global Investment Research

Exhibit 6: There is an influx of new competition at the Application layer; the biggest advantage incumbents have is domain experience

Estimated # of companies and degree of fragmentation by layer in the stack



Source: Pitchbook, Data compiled by Goldman Sachs Global Investment Research

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C. Horizontal eats Vertical — risk score 2

The bear case: Horizontal providers will use their AI-powered tooling to give customers the ability to build vertical-specific workflow within horizontal platforms. The long-term sustainability of the competitive moats that have historically driven significant pricing power for vertical software incumbents comes into question.

GS View: We continue to believe that the characteristics of vertical software, and their respective end-markets, position these businesses to benefit from increasing adoption of AI-technologies, and we see relatively low risk to the sustainability of these competitive moats – assigning a risk score of "2".

The original value case for Vertical Software: While horizontal SaaS applications provide broad levels of functionality across a range of industries, vertical software was created to address industry-specific problems that generic horizontal applications were unable to address effectively; such as unique workflows, regulatory requirements, and/or operational challenges of a particular industry (e.g. GWRE for P&C insurance, VEEV for biopharma, IOT for trucking, PCOR for construction). The cost of specialization, TAM limitations, and often oligopolistic market share dynamics have created a real moat and made these solutions particularly attractive to enterprises requiring more specialized levels of functionality at lower total cost of ownership.

The current defensive moat for Vertical Software: We see four primary advantages inherent in vertical software that provide insulation from horizontal AI new entrants: 1) Access to proprietary, industry specific-data; 2) Centralized modality as a system of record with workflows deeply embedded in daily operations; 3) Longstanding track records of success in select industries, which creates brand referenceability and trust that cannot be manufactured in the short-term; and 4) Regulatory compliance in highly-regulated industries.

1. **Access to proprietary data:** Access to non-public, industry-specific datasets present the most significant barrier insulating vertical software today. This proprietary data is required to train models for high-stakes decision-making in specific industries. For example, ~$775bn of P&C Insurance DWP globally is managed under at least one Guidewire product across the company's customer base of >500 insurers - resulting in a vast amount of historical data that AI-native new entrants cannot access in public forums.
2. **Deep workflow integration supports vertical software as the system of record:** Vertical software is often deeply embedded into the daily operations of the business, making these platforms the mission-critical systems of record in their respective industries. These systems understand the historical patterns and operational characteristics of the business and broader industry, making it increasingly challenging to replace as these companies spend more time operating and innovating.
3. **Brand referenceability cannot be not manufactured in the short term:** While the characteristics of a vertical software business reflect the unique characteristics of its

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particular end-market, the original value case for vertical software broadly is the inherent need for industry-specific solutions. Over time, vertical incumbents have established significant levels of brand referenceability that enable these businesses to continue to gain market share as preferred partners to Enterprises over horizontal SaaS alternatives. While industry-specific characteristics are particularly relevant here, in end-markets such as hospitals & health systems, the public sector, financial institutions, and P&C insurance, brand referenceability is often a key contributor to customer buying decisions.

4. **Compliance requirements in highly-regulated industries:** Complex regulatory environments can act as a barrier preventing AI new-entrants from easily disrupting vertical incumbents. For example, Via Transportation has cited a ~10 month average sales cycle length, in-part driven by the extensive number of regulatory hurdles vendors must clear to participate in public sector RFPs.

Patience has proven to be a virtue in this space: One of the critical questions in the AI / Software thematic debate is how long customers would give an incumbent, embedded software vendor to innovate up to parity with a "better" AI-native solution before moving off of the system altogether. For a vendor with all of its value tied up in UI, minimal embeddedness, and month-to-month contracts, that time might be short. For deeply embedded vertical vendors, that time span can last years (if not longer), and vertical vendors are for the most part not standing still watching others around them lead the innovation conversation - so Vertical leads from a position of strength here. Recall Guidewire's venture into cloud after Duck Creek had a viable cloud product in the market for more than a year in-and-around the late 2010s; Guidewire's on-prem customer base largely avoided knee-jerk switching, and that patience is a large reason why Guidewire is the market leader today.

Horizontal's threat to Vertical Software from rapid advancements in Generative AI: Vertical software has historically relied on solving complex industry-specific challenges that require deep levels of domain-specificity beyond traditional horizontal applications. That said, recent advancements in GenAI have driven rapid levels of innovation, enabling AI natives to enter vertical markets that have historically been dominated by legacy SaaS providers. The expanding capabilities of general-purpose AI models is enabling these companies to perform specialized tasks that have historically required dedicated vertical software applications. Take Palantir, which is working with AIG using Anthropic to assist in a number of Insurance use cases. Similarly, Intuit has launched GenOS to allow its customers to easily code vertical-specific workflows into its horizontal accounting application (Quickbooks). At the same time, vertical industries often have some of the more obvious opportunities for AI applications given so much of the workflow is rules-based. If that piece is displaced by horizontal agents plugging in to existing data, context, and technology, it is an open question what that does to pricing power and profit pools for these companies. We explored this narrative in our Vertical Software launch in January.

Key Question:

- **Is "good enough" functionality enough to pull customers from vertical incumbents?** Historically, we've assumed adoption meaningfully shifts only when AI tools are either clearly superior to incumbent applications and/or materially

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cheaper. However, the built-in advantages of horizontal platforms could still drive migration away from vertical incumbents, even if the AI functionality is merely "good enough" rather than decisively better. Vertical SaaS often reinforces data silos across teams and functions, which can constrain an Enterprise's ability to innovate and scale. By contrast, horizontal applications can offer structural benefits, reducing fragmentation, simplifying integration, and lowering operational complexity - which tends to resonate with Enterprise buyers. A potential inflection point may come when AI-native entrants validate their approach in lower-stakes use cases within traditionally specialized end-markets. Demonstrating credibility in these initial deployments could build confidence in the technology and accelerate broader adoption at scale.

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D. Lower cost of code — risk score 2

The bear case: AI coding tools allow developers to be more productive and enable a wider group of people to develop software, reducing the cost of software development and the barrier to entry.

GS View: *We agree that the cost of code is a key input to the cost of building a software application, and that the cost of producing code is coming down, which will lead to new entrants. However, we currently feel more comfortable about the risk here as there is a substantial difference between coding a product and developing that product into a company: there is more to a successful software company than simply the cost of code. We assign a risk score of "2".*

1. **Software engineering is more than just writing code.** Software engineers already spend a significant portion of their time doing non-coding tasks, including higher level design thinking, fixing bugs, identifying risks, and reviewing other engineers' work. Advances in coding have until now not diminished the need for engineers. 20 years ago, coders had to write their own memory management functions, data serializers, HTTP requests, etc., and today these are handled for the engineer as programming languages and frameworks abstract these details. But the work of a software engineer, and the companies who employed them, did not go away. It is true that vibe coding tools are advancing at a substantial rate, and increasingly do the job of proofing code and designing UIs as well as humans can. But that does not take away the other parts of the development life cycle.
2. **Software companies are more than just an aggregation of code.** There are >15,000 SaaS companies in the US alone, but very few have been successful at scale. Success involves more than just an interface; data, task accuracy, security, maintenance, workflow orchestration, having others build on top of the platform, integrating into other vendors, and go-to-market. **Ultimately, we believe the risk lies more in bespoke software (discussed below) than in vibe coding** because it acknowledges the broader complexity that goes into creating the software stack.
3. **The code that gets generated likely still requires humans in the loop.** Coding tools are progressing to the point where the quality of the code output is improving significantly. That said, we believe there will still need to be humans in the loop to understand, proof, and maintain code - and that faster code generation could actually pull forward roadmaps. Take Faros' study of 10,000 developers mid-last year: Teams with high AI adoption completed 21% more tasks and merged 98% more pull requests, but pull request review-time increased 91%. There are a whole group of DevOps companies that help Engineers with these AI-specific bottlenecks, but there is no "one size fits all" answer to skipping past a complex AI generated code lifecycle, particularly in an Enterprise setting.

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E. The future of software is bespoke — risk score 3

The bear case: As the cost of code goes down, more enterprises will choose to build their own apps that are tailored to specific workflows, data, and objectives, instead of buying software from software vendors. This opportunity either accrues to the infrastructure software vendors (such as Databricks and Snowflake), LLM model providers, or Palantir.

GS View: We do not believe the cost of code coming down changes the calculus universally on build vs. buy, although we do expect to see some share capture by custom software in the enterprise. In our view, this is less a question about vibe coding and more a question on internal IT choosing to build enterprise-grade software in house because they believe they can do so at lower cost for better performance. This distills down to the classic question of insourcing vs. outsourcing: while the cost of building the software may go down, the cost and responsibility of maintaining it (even with agentic efficiencies) will still compound over time. Even in a scenario where the cost of maintaining software applications internally goes down, it will also go down at the specialized providers, such that the performance to cost frontier at the specialized provider will generally always be ahead of the in house technology. We see the most likely custom software share capture in the space that falls between typical systems of record (back office vs. front office, and requires coordination across multiple business units or teams that may not have been otherwise connected by packaged software). We assign a risk score of "3".

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Historically, SaaS vendors created value by bundling user interfaces, workflow logic, permissions, reporting, and data models into a single, standardized product. This model worked because it centralized R&D: vendors could amortize the cost of specialized engineering talent, compliance, scalability, and reliability across a large customer base. For buyers, this made purchasing software cheaper, faster, and operationally simpler than building in-house solutions, while reducing implementation risk and ensuring ongoing upgrades without the need for large, permanent development teams. As a result, packaged SaaS became the dominant delivery model across most enterprise application categories.

AI may challenge this dominance:

- **Disruption in software historically requires solutions that are both meaningfully better and meaningfully cheaper.** As internal teams are able to ship code faster, iterate continuously, and automate portions of development and maintenance, custom-built applications become more viable, particularly when they optimize for company-specific processes rather than horizontally standardized workflows designed to be "good enough" for a broad customer base. Value increasingly accrues to systems that are tightly coupled to an enterprise's proprietary data and operating logic. This shift does not eliminate the need for platforms, but it reduces the differentiation of application-layer software whose primary value proposition is standardization.
- **Palantir represents the clearest example of this model in practice.** Palantir's approach centers on bespoke, co-developed applications built directly on top of customer data and workflows. Its ontology enables data integration and custom operational logic, but the increased enterprise focus on AI has accelerated adoption

of this model. Through its AI Platform (AIP), Palantir has demonstrated the ability to deliver production-grade AI use cases with relatively fast time to value and measurable ROI, which has translated into an outsized share of enterprise AI budgets and best-in-class growth rates (Palantir's US Commercial business grew 109% in 2025 and the company expects to accelerate to over 115% in 2026E).

- One example is a supply chain optimization application developed jointly by Palantir and a beverge customer's engineering team, in which AI agents manage delivery, shipping, and inventory decisions across tens of thousands of loads and containers. According to the customer, the system was built in months rather than years, illustrating how AI-enabled development can materially compress deployment timelines for highly customized, mission-critical applications.

- **Palantir's success is partly a function of its use of forward deployed engineers (FDEs).** These FDEs embed directly with customers to translate siloed data and tacit operational knowledge into product-grade bespoke applications. This allows customers to move from prototype to enterprise-wide deployment, converting customer-specific solutions into reusable platform capabilities that scale across similar operating environments.
  - While investors historically viewed this model skeptically (i.e. is this software or services?), Palantir has achieved gross margins of ~85% on its blended business model, and FDEs are increasingly adopted among all cohorts of new competition. FDEs serve as the translation layer between intent and machine autonomy, reducing risk while enabling scalable, real-world deployment. The Palantir FDE model leverages platform verticalization to realize leverage on an otherwise services-intense business: by pairing FDEs with a separate product development team, Palantir converts customer-specific applications (from the FDEs) into reusable platform capabilities (from the product development team).

We believe we may be near a local maximum on the number of enterprises choosing to build bespoke as the packaged software ecosystem is maturing. SaaS incumbents are only now beginning to offer comprehensive AI functionality and IT professionals are still early in their learning curves of how to make AI projects successful. At the same time, organizations are having to face decades of sub-optimal data management practices, and compounding security and governance challenges associated with building software on poorly organized data. As a result, building custom solutions currently seems like the best way to derive value from AI. However, we think these challenges are being managed: SaaS incumbents are continuing to expand AI functionality, developers are becoming more proficient, data strategy has been in focus over the last 12-18 months, and security and governance protcols (A2A, MCP, etc.) are evolving. ServiceNow has recently discussed winning share of budget that had previously been allocated to bespoke builds.

- Many enterprises continue to value standardized applications that embed regulatory compliance, security, uptime, and cross-industry best practices, particularly in regulated or mission-critical domains.

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- Custom-built systems shift long-term ownership of maintenance, governance, and reliability back to the customer, which can offset upfront development savings.
- In parallel, leading SaaS incumbents are embedding AI agents and extensibility directly into existing platforms, preserving distribution, data gravity, and switching costs while enabling greater workflow flexibility; as discussed above, incumbents are also integrating FDEs into software distribution efforts in order to expand customization.

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F. Gross margin pressure & the "LLM Tax" — risk score 3

The bear case: Software's high gross margins (70–90%) are structurally unsustainable in an AI-centric landscape. As competition goes up, new entrants may deliver comparable or superior outcomes with lower company cost structures (having started with a clean sheet of paper and no incumbent profit pool to protect). At the same time, the marginal cost of production at existing providers goes up from 10-20% (the cost of CPU-based hosting) to something greater (the cost of GPU-based inference).

GS View: We expect a 12-24 month period of modest gross margin pressure in aggregate, as many companies will focus on customer adoption before monetization and choose to absorb the cost of GPU inference and LLM APIs. However, for the industry, inference costs will likely continue to decline over time relative to the value/amount of associated productivity of AI. For any specific company, gross margin is ultimately a function of pricing power, which in turn is a function of product differentiation (where incumbents could have an edge if they convert their domain experience into higher quality outcomes). Our investment process will favor those companies that can show product differentiation. Given uneven outcomes across companies, we assign a risk score of "3".

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Historically, SaaS vendors have realized gross margins of 70-90% with a business model that is fundamentally asset light and IP-driven. While companies incur significant upfront R&D expenses to build the software, once it is deployed, the marginal cost of serving an incremental user is tied to the associated cost of CPU compute (typically less than 20% of revenue). As scale increases, SaaS vendors also benefit from unit cost declines in compute, storage, and bandwidth. At the same time, many SaaS vendors had pricing power driven by high switching costs, workflow and data entrenchment, and limited competitive substitutes. This enabled software companies to upsell and cross-sell, which allowed revenue per customer to compound faster than infrastructure costs and support gross margin expansion over time.

Software gross margins today are beginning to see pressure from higher variable costs and new competition:

- **There are two new costs for AI functionality: the cost of GPU inference, and the cost of using a third party model provider.** Unlike traditional SaaS models where the marginal cost of incremental usage was effectively de minimis, AI-enabled features convert engagement directly into expense, with token consumption, model complexity, and query frequency directly driving costs. As AI adoption scales, product usage is no longer purely gross profit accretive; it now carries a compute burden that grows with interaction intensity and LLM model pricing. This marks a structural shift in the SaaS model, from fixed-cost IP leverage toward consumption-based economics, with COGS increasingly tied to user behavior and third-party cost curves.
    - This is particularly evident among private AI startups. According to Bessemer Venture Partners, the fastest-growing AI-native startups (i.e., scaling from $0 to $100M ARR within two years) are operating at roughly ~25% gross margins, with many others reporting negative margins as they prioritize distribution over near-term profitability. More mature AI-native companies that are

scaling more deliberately tend to exhibit gross margins closer to ~60%, still below traditional SaaS.

- **That said, inference costs should decline over time with advances in model efficiency, hardware performance, routing between models, and competitive dynamics among AI providers (see our Semis team's work on the inference cost curve).** While varying by benchmark and tasks, LLM inference costs have declined by 9x to 900x per year (Epoch AI). For example, the price to achieve GPT 4's performance with similar MMLU score has declined by 40x per year. Next generation models today are designed in a more efficient manner with improving token efficiency. Today, the performance benefit of new models is passed to end users in most instances due to competition. In the future, companies may balance between passing cost downs to the end user and delivering margins to shareholders. There are two longer term swing factors:
  - For LLM costs: we believe the key swing factor will be the degree to which LLMs are differentiated vs. commoditized.
  - For compute costs: Recall that Microsoft expects its underlying gross margins in AI-based Azure to ultimately converge up to CPU-based Azure - in this scenario, the cost of compute to Software vendors that use hyperscalers/neoclouds would also converge to CPU-based economics. Software vendors can also mitigate inference costs through batching, caching, model fine-tuning, and selectively deploying lower-compute models for simpler use cases. If inference costs fall faster than AI-driven usage scales, the incremental COGS burden could compress naturally, suggesting that margin pressure may prove transitional rather than structural.
- **Some are better positioned than others on the "LLM tax" challenge.** With a vertically integrated stack and a strategic relationship with OpenAI, Microsoft is able to internalize model economics, optimize inference costs, and capture margin across multiple layers of the value chain rather than absorbing third-party "LLM tax" leakage. Currently, Microsoft having IP rights to OpenAI models has allowed the company to incur no gross margin tax from API calls, which we view as a competitive advantage vs. other software providers.

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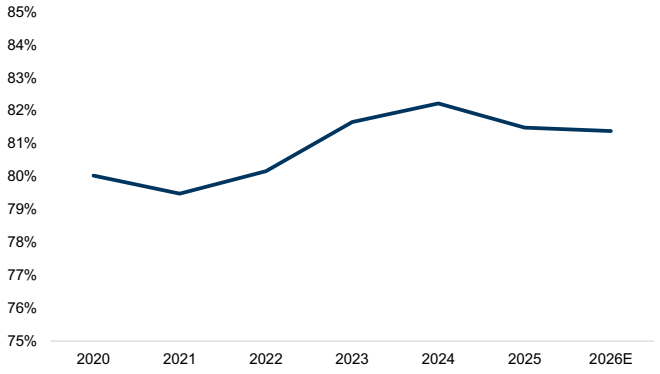
Exhibit 7: There is likely some medium term gross margin risk to Street estimates, but longer term gross margin will be a function of pricing power, in turn a function of product differentiation
Average gross margins for Salesforce, ServiceNow, Workday, and HubSpot from 2020-2026E



Source: Company data, Visible Alpha Consensus Data, Goldman Sachs Global Investment Research

The question on whether gross margin compression is temporary or structural also requires looking beyond LLM model economics to the core question of pricing power and product differentiation:

- **Historically, software has captured ~10–25% of the value that it creates.** While AI is expanding the TAM, the primary variable to value capture will be the degree of new competition. Software may still benefit from TAM expansion but this benefit may be more than offset by a structurally smaller share of the profit pool, resulting in sustained pressure on pricing power and margins over time.
- **Maintaining software margins ultimately hinges on pricing power which hinges upon product differentiation.** Vendors that price at high gross margins must consistently demonstrate clear, differentiated value to justify that value capture; otherwise, customers will migrate as soon as credible alternatives emerge. The next phase of the market will test which software providers are delivering truly indispensable value versus those benefiting from legacy pricing structures.
- **We believe software companies are prioritizing adoption over monetization** by bundling AI features into existing products, offering free/lower priced tiers, and optimizing for usage rather than immediate revenue. For example, Figma included a headwind to gross margin in its NTM forecasts as Figma Make (its prototyping product) scales, and ServiceNow includes an initial tranche of free AI credits in some of its bundles. The goal is to maximize distribution and customer lock-in before pricing fully reflects value, especially while customers are still in early-stages of deploying AI at scale and thus recognizing its value. This approach pressures near-term gross margins due to higher inference and partner costs that are not yet fully passed through. However, this is likely a transitory phase, and margins should improve as usage-based pricing models mature, costs decline, and AI is more fully deployed and thus driving value.
- **Another positive offset is software potentially taking share from the services**

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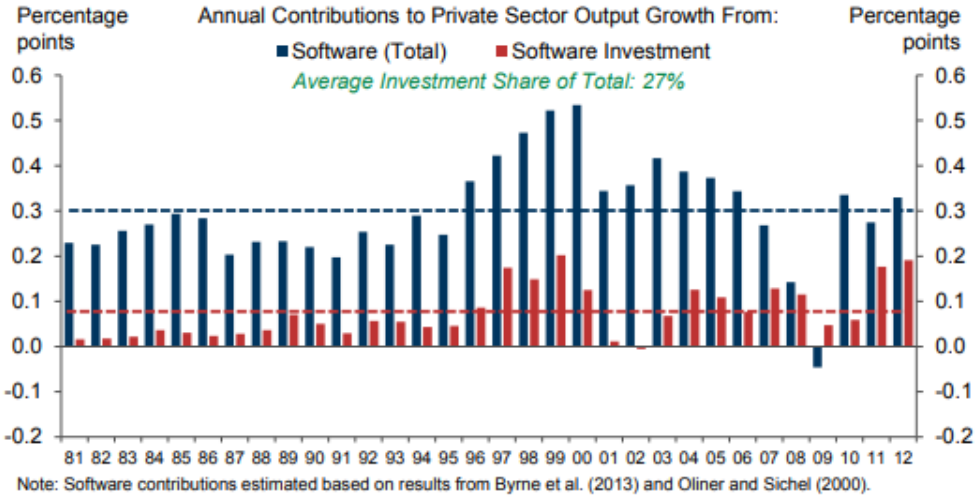
and labor pool. Productivity gains increasingly come from substituting automated software for human expertise, allowing vendors to monetize savings previously allocated to labor. Intuit's assisted tax solutions illustrate how spend can shift away from human tax professionals toward software platforms. Similarly, in advertising, AI-powered tools have the potential to abstract and automate functions historically delivered by ad agencies, reallocating profit pools away from services and toward digital channels through lower costs and reinvestment into ad spend (see our internet team's deep dive). In both cases, AI compresses the need for manual expertise, strengthening substitution dynamics. This suggests that software vendors with differentiated AI-driven outcomes may sustain value capture (and in turn gross margin) by taking share from labor and services budgets.

Exhibit 8: ~25% of private sector growth gains were captured by Software companies between 1981 and 2012



Source: Byrne et al. (2013), Oliner and Sichel (2000) , Federal Reserve Board, Goldman Sachs Global Investment Research

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G. The pace of innovation increases uncertainty — risk score 5

The bear case: The technology is rapidly changing, meaning the end state is difficult if not impossible to predict. Scaling laws still hold. YTD, there have been notable updates from Anthropic (Cowork, Opus 4.6, vertical plug ins), OpenAI (Frontier, OpenClaw), Google DeepMind (Deep Think) and Meta (Avocado). This creates substantial uncertainty about future knowledge work, the economic structure of the Software industry, and the terminal value of the companies - and uncertainty is a low multiple business.

GS View: *This risk is the hardest to plan for, and our due diligence continues to include leading edge AI startups and AI academics. However, uncertainty could also surface new market opportunities: consider the potential for TAM creation in medical diagnostics or energy, as Microsoft discusses in its superintelligence blog. We assign a risk score of 5.*

AI is both continuing to advance in terms of power/performance *and* advancing in new directions:

- **Scaling laws are holding; state-of-the-art continues to improve.** Bridgewater's November white paper showed that scaling laws for AI pre-training continue to hold: Gemini 3 Pro at that point in time was the latest example of a model that was larger in size compared to its predecessors (GPT-5.1 and Claude Opus 4.5) showing improvements across visual, text, and coding capabilities. Since then OpenAI released GPT-5.2 on 12/11/2025 and Anthropic released Claude Opus 4.6 on 2/5/2026, each improving on prior models. Some benchmarks are nearly solved: GPT-5.2, Gemini 3 Pro and Opus 4.6 all score >90% on graduate-level reasoning as measured by GPQA Diamond. But there remains room for improvement on others: Claude Opus 4.6 scores 53.1 on the Humanity's Last Exam (HLE) benchmark vs. GPT-5.2 Pro 50.0. There may be an upper bound to model pre-training but research labs have not yet hit it. The hardware also continues to improve, meaning more powerful models trained on e.g. Rubin have yet to be benchmarked.
- **New applications of the existing technology also drive innovation.** Innovation is not just about pushing the state-of-the-art, it can also mean leveraging existing technology in new ways. Examples:
  - **ChatGPT:** Breakthroughs in natural language processing began in the late 2010s and in 2020, OpenAI showed with GPT-3 that scaling up model size led to highly capable language understanding. What brought LLMs into the mainstream and drove a breakout moment in December 2022-January 2023 was *packaging*. ChatGPT was easy to access and use. This allowed the more people to use it which enabled broader adoption, which in turn spurred further experimentation. One potential reason why OpenAI released ChatGPT to a general audience was that beta testers didn't know what to talk to ChatGPT about (link). Once a wider audience started experimenting with the technology, a wider array of use cases emerged.
  - **Claude Cowork:** In our industry conversations, developers had already been leveraging Claude Code and the feedback on the technology was very positive.

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Many were even surprised at the increased interest around Claude Cowork given they had already been working with the underlying technology of Claude Code. What changed with Claude Cowork was the rollout to the Claude Desktop app which provided a GUI that let non-technical users experiment with the technology. Cowork will likley continue to find new applications, which makes the end state of the technology difficult to predict.

- **OpenClaw** (formery Clawdbot/Moltbot): Last week in our conversation with Matthew Prince (co-founder/CEO of Cloudflare), Mr. Prince noted that OpenClaw's progression in the next 3 years could mirror ChatGPT's progression over the last three. The idea of self-hosting an AI agent is not, per our industry conversations, all that technically difficult. The main gating factor to effectiveness is the user's willingness to sacrifice security. Consumers willing to compromise on security have been experimenting with the technology and creating novel tools and workflows and we believe this will continue - particularly as an ecosystems of tools emerge to help solve usability, security and technical challenges. Enterprise use cases will likely be limited near term due to the security issues but if a way to secure self-hosted agents based on OpenClaw is established, this would open new directions in which the technology can go.

- **Uncertainty could create new opportunities.** For example, **in November 2025, Microsoft announced its MAI Superintelligence Team within Microsoft AI.** Its blog cites use cases including: 1) Medical superintelligence to provide expert level performance across diagnostics, and planning and prediction in operational clinical settings. A key datapoint here is from the MAI-DxO model. The New England Journal of Medicine includes a Case Challenge in every issue – a list of symptoms and a patient to diagnose. Pass rates are in the low single digit percentages even for domain experts. MAI-DxO has achieved an 85% pass rate across the Case Challenges. Inputting Microsoft's blog post and its initial metrics of success into ChatGPT yields TAM calculations of $50-$100bn per year, with an upside case of $150-200bn, which in turn is consistent with generic AI in healthcare forecasts. **Gemini 3 Deep Think released an update in February** for advancing science, research and engineering. Our favorite example cited in the release is a science lab at Duke University that used Deep Think for a semiconductor fab application, formulating a receipt to grow thin films larger than 100 μm, meeting a precise milestone that previous methods fell short of.

- **Unknown unknowns:** Some technological problems are known to us now, and we can speculate as to their evolutions. For example, scaling and efficiency laws can help us make predictions about model capabilities or token usage 1-2 years out, and there is already discussion in the academic world as to whether LLMs can truly make progress toward AGI and alternative neural network approaches. These are the *known unknowns*. But not all advances are immediately obvious; there will be breakthroughs that fundamentally alter what is possible, and consequently the end state of the industry. We are still arguably living in the ChatGPT era, where for most people AI means interactions with intelligent chatbots (Exhibit 10). This could change in ways that are impossible to predict, which in turn would have implications

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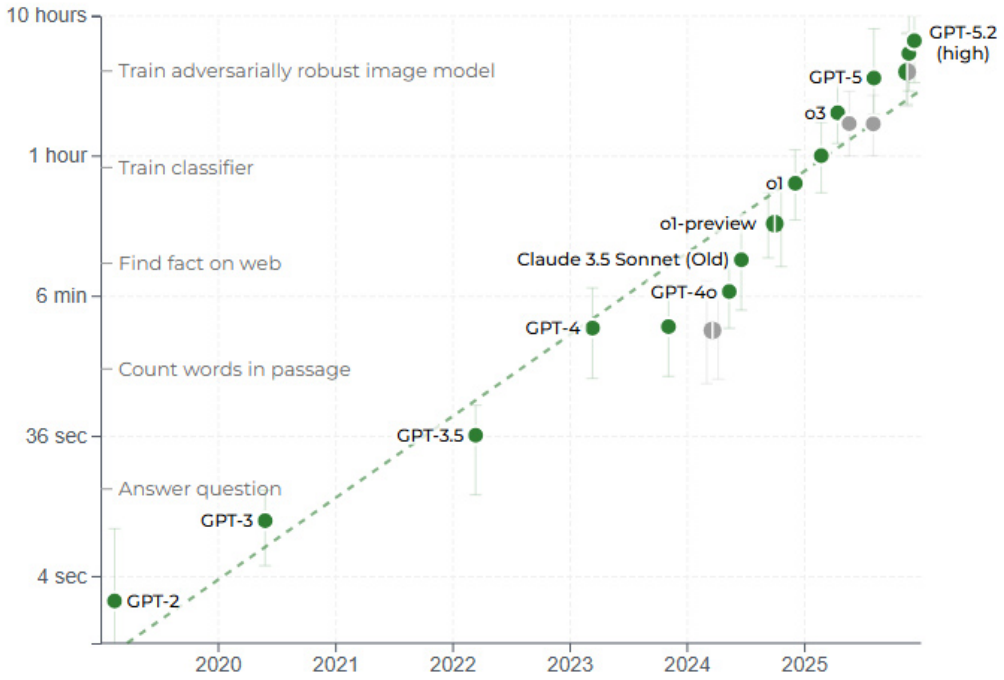
for the companies the technology touches. An observer in 1993 would have found it difficult to predict what Web 2.0 would look like, and few observers in December 2022 might have predicted Claude Cowork/Clawdbot and their impacts on the industry narrative. Such *unknown unknowns* make it impossible to accurately predict the future state of artificial intelligence. This does not mean that negative outcomes are inevitable (or even likely), but it does make them harder to rule out in the near term. This drives uncertainty.

Exhibit 9: AI models continue to follow scaling laws, advancing in ways that open up new applications
Amount of time needed to perform select tasks with 50% accuracy, log scale



Source: METR

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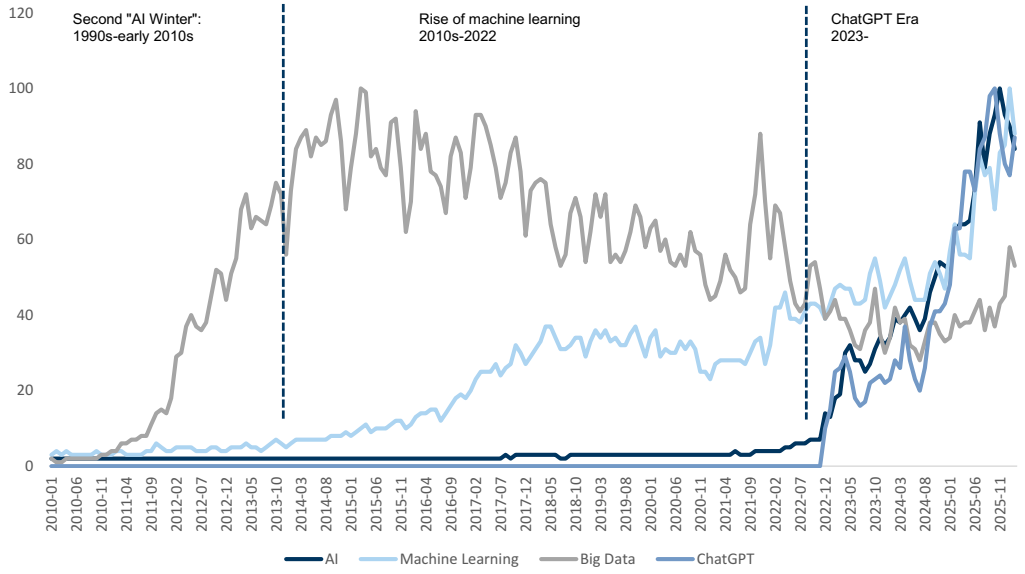
Exhibit 10: But we are still in the ChatGPT era of AI as far as the majority of use cases are concerned, based on Google Search data
100=Peak search interest



Source: Google Trends (https://www.google.com/trends)

Exhibit 11: Unknowables carry major implications and uncertainty about the end state of the technology and the industry

	Questions	Implications	Uncertainty
Known Knowns:	What are the companies' core competencies and market shares today? What are the near term demand signals?	NTM growth Nearterm product roadmaps	**Small**
Known Unknowns:	What will the macroeconomic environment be in 2 years? What will be the cost of compute in 2 years? Will scaling laws hold?	Next 3 years of growth, margins Improvements on the bleeding edge	**Medium**
Unknown Unknowns:	Impossible to know; examples include: How will AI be produced and consumed in 10 years? What will its applications and role in society be?	Birth of new subsectors of software Successful companies that may have not been founded yet	**High**

Source: Goldman Sachs Global Investment Research

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We, Gabriela Borges, CFA, Adam Hotchkiss, Matthew Martino, Noah Naparst, Maura Hager, Greyson Sklba, Callie Valenti, Selina Zhang, Max Gamperl and Praachi Arora, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

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